Sample questions for CO1 – Assessment Tool 3 – Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.
6. What are web services? Explain how web services facilitate communication between cloud applications.
7. Discuss the major advantages and challenges of cloud computing. Explain how organizations can benefit from cloud adoption.

ASSESSMENT TOOL

$CO_1 - AT_1$.

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1. Cloud Computing.

* Cloud computing is a computing resource that provide servers, networking, database, software through internet.

The four characteristics that make it different from traditional on-premise computing

⇒ On demand self service:

Resources can be used whenever they need without any human involvement

⇒ Broad network access:

Resources can be used through Internet by mobile, laptop etc anywhere

Example : Google cloud.

⇒ Resource pooling :

Many Physical server is divided into multiple virtualization server.

Rapid elasticity.

Rapid increase or decrease base on the demand.

Example: Great Indian Amazon prime.

Tradational on premise computing	cloud computing.
⇒ Hardware ⇒ Not easily accessed ⇒ cost is High	⇒ Software base ⇒ Easily afortable ⇒ Low cost

2. Evoloution of cloud computing.

Cloud computing is evoloud through several different phases

* Hardware
* Personal computer
* Network
* Internet or Software
* Virtualization
* cloud computing.

These six are the different phases of cloud computing.

Hardware.

Large frame computers

Personal computer

Individual use their own PC's

Network.

Computer are connected through LAN and VAN.

Internet.

The connection is made through internet among worldwide

Virtualization

One physical server is divided into many virtualization server.

Example * Windows VM

* Linux VM

* Ubuntu VM

cloud computing

It is a computing resource that provide servers, network, database through Internet.

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3). Server virtualization in cloud computing.

Server virtualization is the process of dividing the physical server into many virtualizalization server using hypervisor.

It provides

=> Own CPU

=> Own RAM

=> Own server and

=> Own database.

Difference between server virtualization as an enabling technology and cloud computing as a service model.

Virtualization	cloud computing
=> Technolog that creats virtual resource	=> It provides resource over the Internet
=> Acts as enabling technology	=> Provides on-demand
=> There as several Virtual boxes * Windows VM * Linux VM * Ubuntu VM	=> Software acts as service through Internet.

5. Differentiate IaaS, PaaS, SaaS, and XaaS.

IaaS => Infrastructure acts as service

Provides => Virtual machine, storage network

Example => AWS EC2.

PaaS => Platform acts as Service

Provides => Development platform, runtime tools

Example => Google App

SaaS => Software acts as Service

Provides => Ready to use software

Example => Google. Doc

XaaS => Anything acts as Service

Provides => Various IT resource as service

Example => AWS service.

4. Role of Data Center.

A data center is a building that contain thousands of servers, storage device, networking equiments and power supply cooling system

1). Store data.

It store the data of the user personal as well as professional.

2). Runs Application.

The application such as amazon and netflix

3) Security.

It provides better security such as ⇒ Firewall

⇒ Encryption

⇒ Access control.

4). Backup and disaster recovery.

Multiple copies of data are taken and store in multiple places so that if disaster accurs in one place other data are removed so backup is necessary.